Algorithm: Library Management System

1.Start the program.

2.Create an empty list named books.

3.Display the library menu:

- Add Book
- View Books
- Search Book
- Remove Book
- Exit

4.Read the user’s choice.

5.If the choice is 1, input a book name and add it to the list.

6.If the choice is 2, display all books in the library.

7.If the library is empty, display “Library is empty.”

8.If the choice is 3, input a book name to search.

9.If the book exists in the list, display “Book is available in the library.”

10.Otherwise display “Book not found.”

11.If the choice is 4, input a book name to remove.

12.If the book exists, remove it from the list.

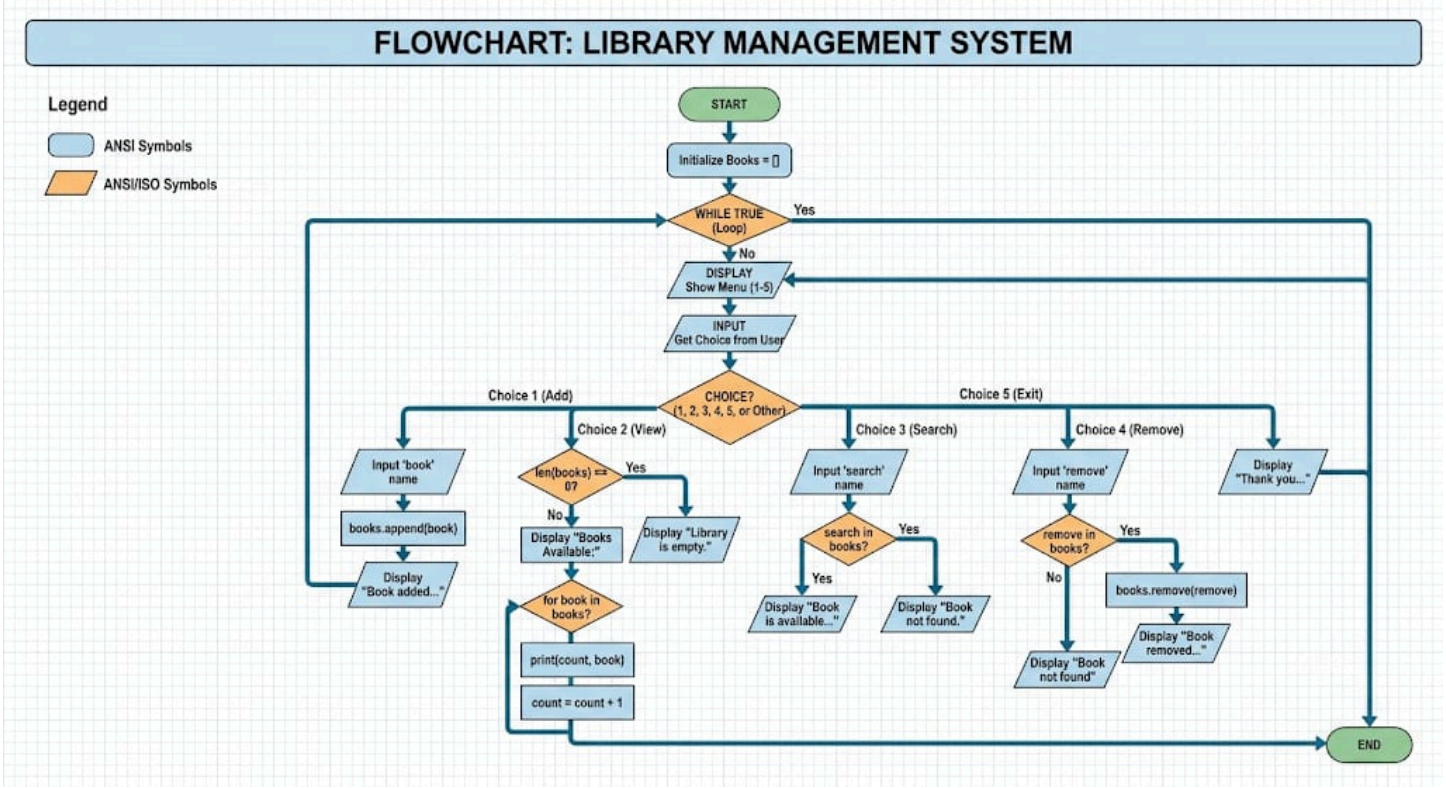
13.Otherwise display “Book not found.”

14.If the choice is 5, display a thank-you message and terminate the program.

15.For any other choice, display “Invalid choice.”

16.Repeat steps 3–15 until the user exits.

17.End the program.



Strategy Pattern Implementation - Python Source Code

```
books = []


while True:
    print("\n===== Library Management System =====")
    print("1. Add Book")
    print("2. View Books")
    print("3. Search Book")
    print("4. Remove Book")
    print("5. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        book = input("Enter book name: ")
        books.append(book)
```

```
    print("Book added successfully!")

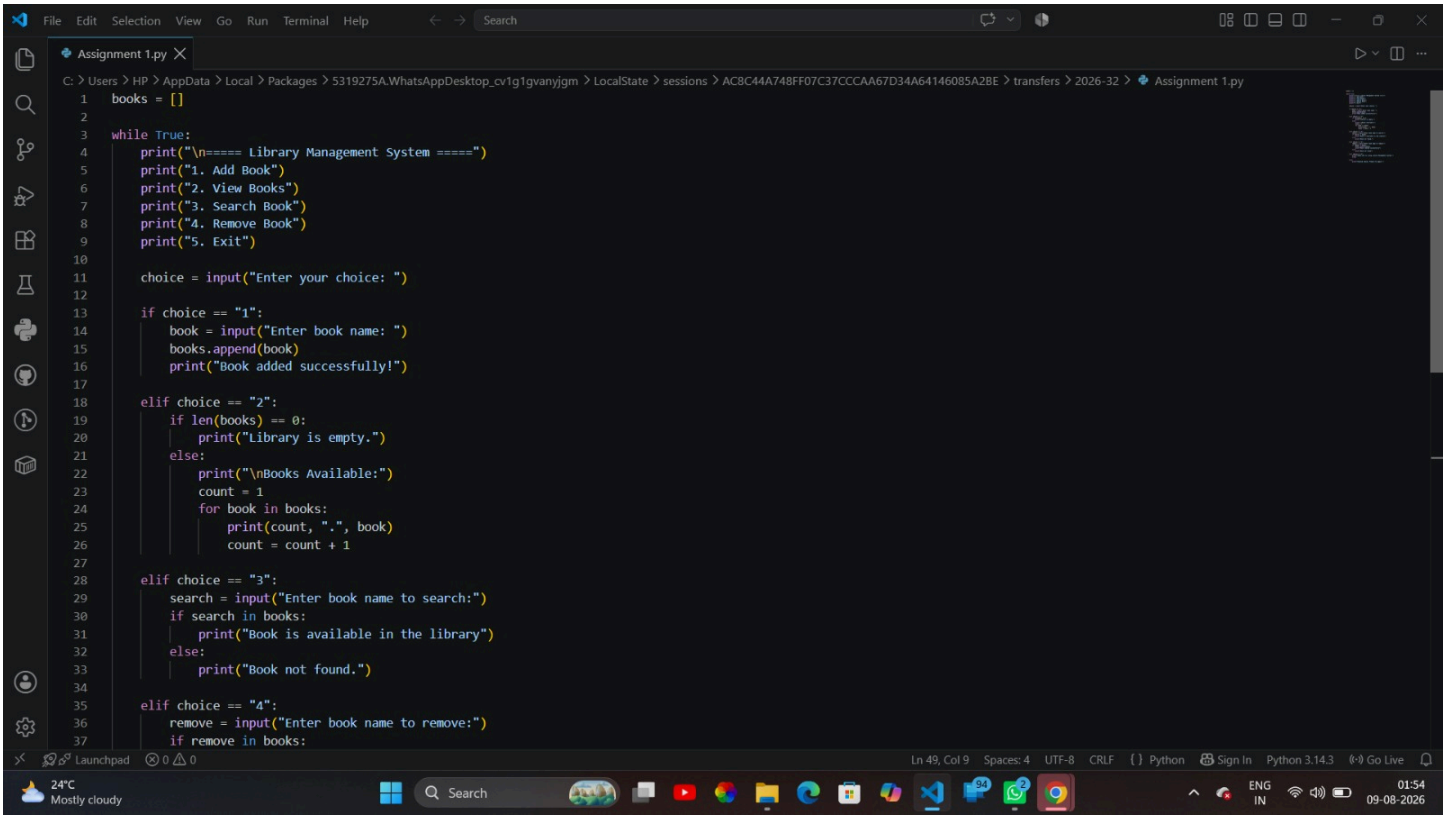
elif choice == "2":
    if len(books) == 0:
        print("Library is empty.")
    else:
        print("\nBooks Available:")
        count = 1
        for book in books:
            print(count, ".", book)
            count = count + 1

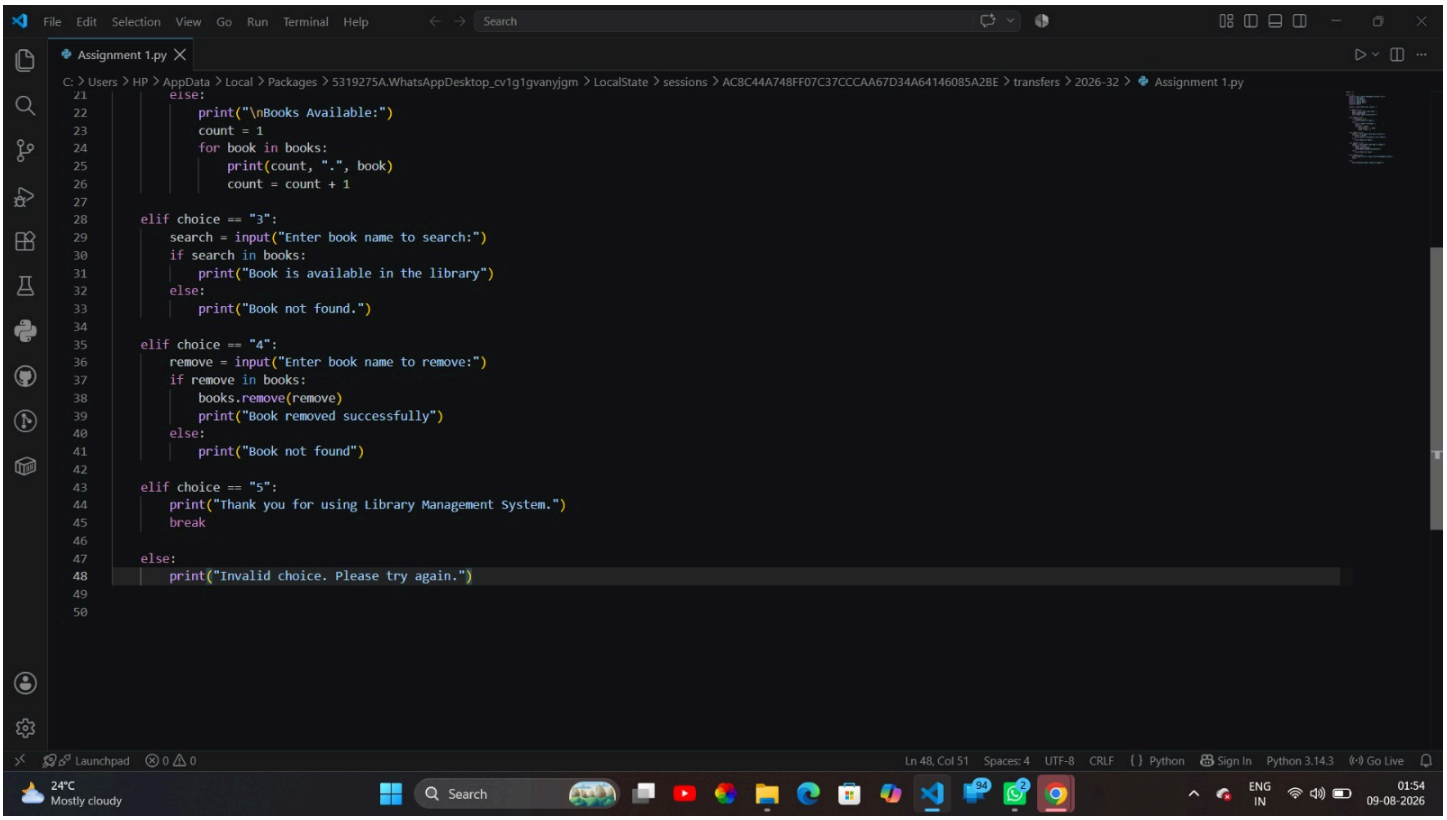
elif choice == "3":
    search = input("Enter book name to search:")
    if search in books:
        print("Book is available in the library")
    else:
        print("Book not found.")

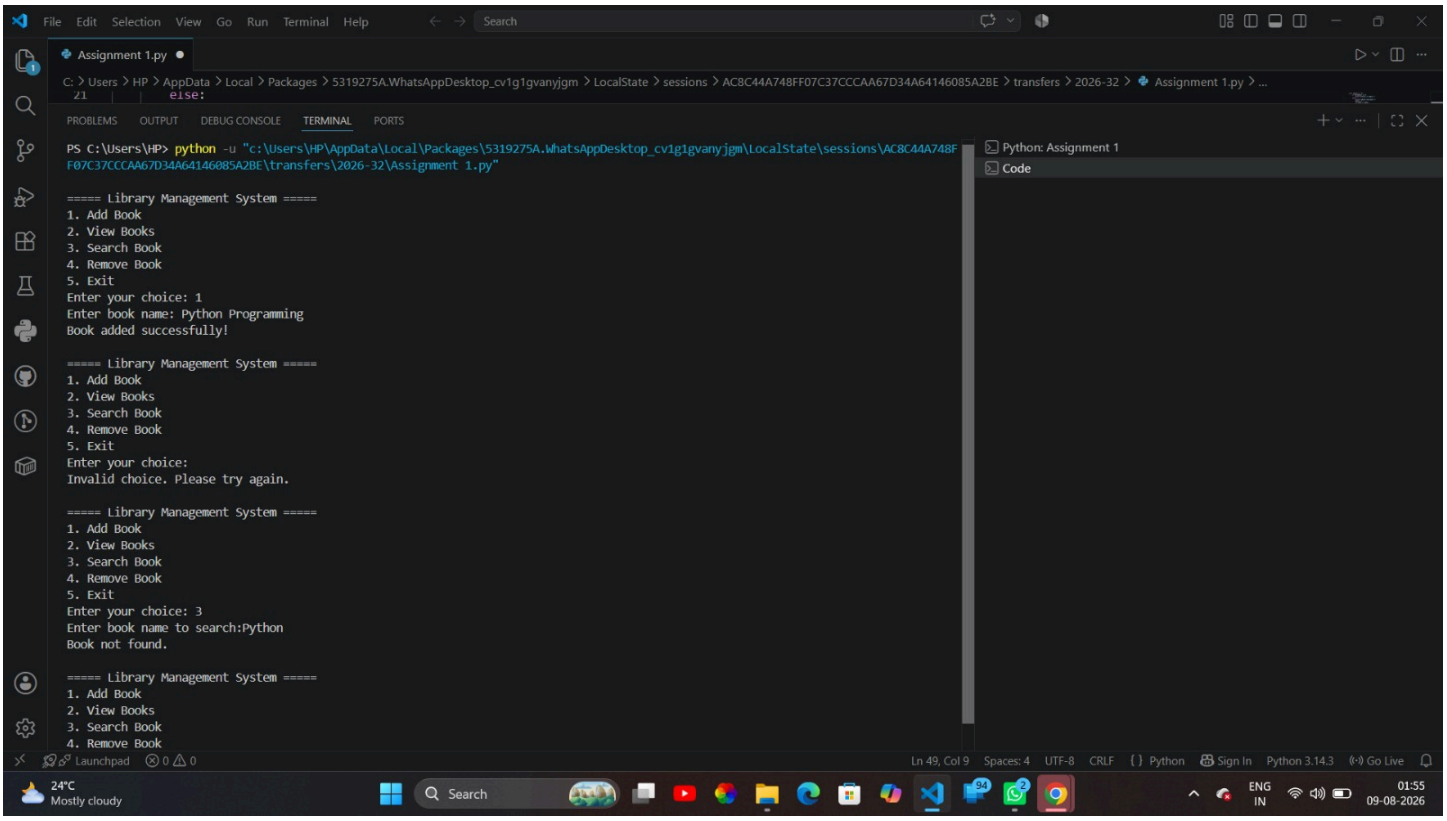
elif choice == "4":
    remove = input("Enter book name to remove:")
    if remove in books:
        books.remove(remove)
        print("Book removed successfully")
    else:
        print("Book not found")
```

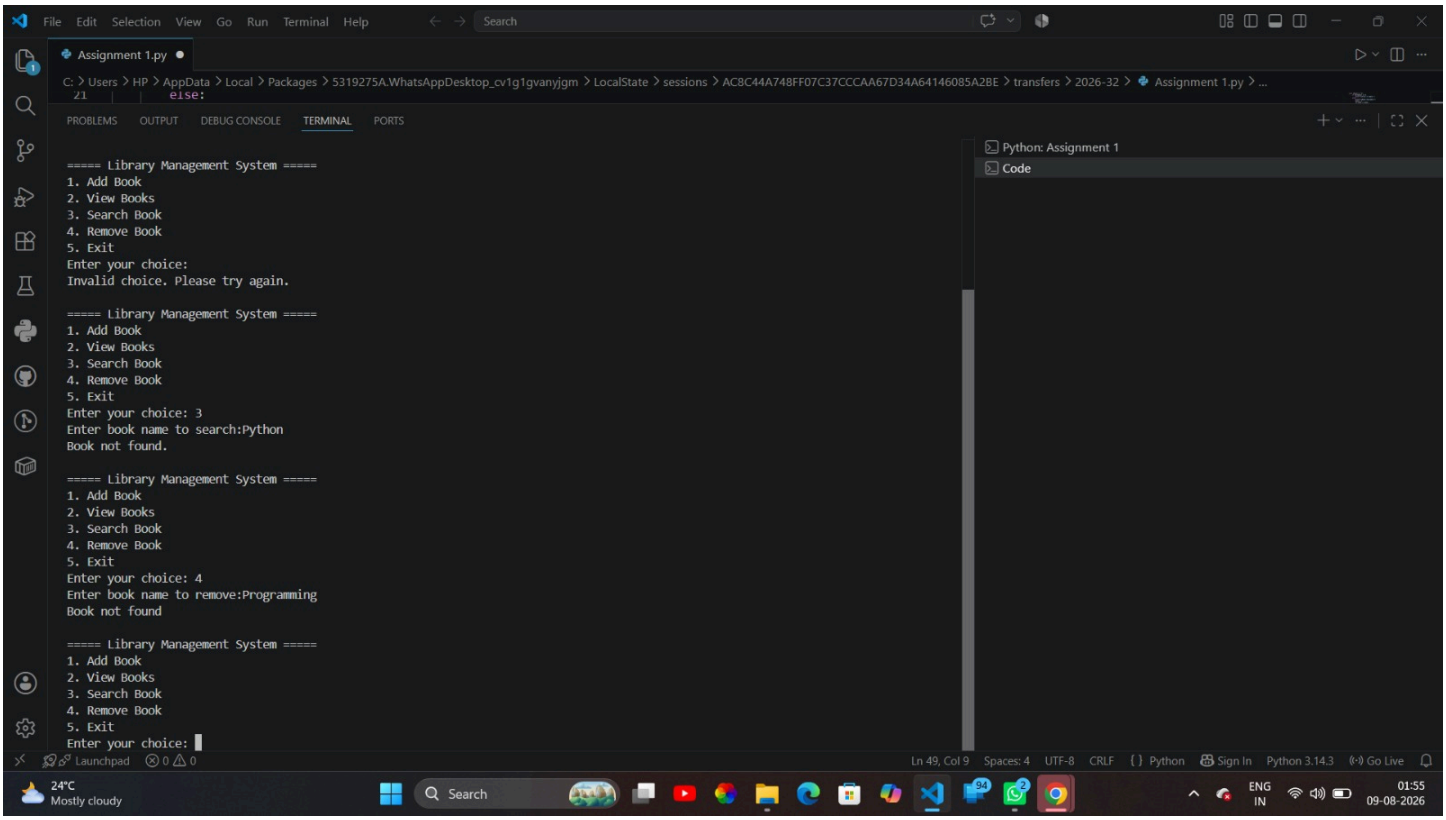
```
elif choice == "5":
    print("Thank you for using Library Management System.")
    break

else:
    print("Invalid choice. Please try again.")
```




File Edit Selection View Go Run Terminal Help
Search
Assignment 1.py
C: > Users > HP > AppData > Local > Packages > 5319275A.WhatsAppDesktop_cv1g1gvanyjgm > LocalState > sessions > AC8C44A748FF07C37CCCAA67D34A64146085A2BE > transfers > 2026-32 > Assignment 1.py
else:
print("\nBooks Available:")
count = 1
for book in books:
print(count, ".", book)
count = count + 1
elif choice == "3":
search = input("Enter book name to search:")
if search in books:
print("Book is available in the library")
else:
print("Book not found.")
elif choice == "4":
remove = input("Enter book name to remove:")
if remove in books:
books.remove(remove)
print("Book removed successfully")
else:
print("Book not found")
elif choice == "5":
print("Thank you for using Library Management System.")
break
else:
print("Invalid choice. Please try again.")
Launchpad
Ln 48, Col 51
Spaces: 4
UTF-8
CRLF
Python
Sign In
Python 3.14.3
Go Live
24°C
Mostly cloudy
Search
ENG
IN
01:54
09-08-2026


File Edit Selection View Go Run Terminal Help
Search
Assignment 1.py
C: > Users > HP > AppData > Local > Packages > 5319275A.WhatsAppDesktop_cv1g1gvanyjgm > LocalState > sessions > AC8C44A748FF07C37CCCAA67D34A64146085A2BE > transfers > 2026-32 > Assignment 1.py > ...
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS C:\Users\HP> python -u "c:\Users\HP\AppData\Local\Packages\5319275A.WhatsAppDesktop_cv1g1gvanyjgm\LocalState\sessions\AC8C44A748FF07C37CCCAA67D34A64146085A2BE\transfers\2026-32\Assignment 1.py"
===== Library Management System =====
1. Add Book
2. View Books
3. Search Book
4. Remove Book
5. Exit
Enter your choice: 1
Enter book name: Python Programming
Book added successfully!
===== Library Management System =====
1. Add Book
2. View Books
3. Search Book
4. Remove Book
5. Exit
Enter your choice:
Invalid choice. Please try again.
===== Library Management System =====
1. Add Book
2. View Books
3. Search Book
4. Remove Book
5. Exit
Enter your choice: 3
Enter book name to search:Python
Book not found.
===== Library Management System =====
1. Add Book
2. View Books
3. Search Book
4. Remove Book
Python: Assignment 1
Code
Launchpad
Ln 49, Col 9
Spaces: 4
UTF-8
CRLF
Python
Sign In
Python 3.14.3
Go Live
24°C
Mostly cloudy
Search
ENG
IN
01:55
09-08-2026


File Edit Selection View Go Run Terminal Help
Search
Assignment 1.py
C: > Users > HP > AppData > Local > Packages > 5319275A.WhatsAppDesktop_cv1g1gvanyjgm > LocalState > sessions > AC8C44A748FF07C37CCCAA67D34A64146085A2BE > transfers > 2026-32 > Assignment 1.py > ...
21 else:
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python: Assignment 1
Code
===== Library Management System =====
1. Add Book
2. View Books
3. Search Book
4. Remove Book
5. Exit
Enter your choice:
Invalid choice. Please try again.
===== Library Management System =====
1. Add Book
2. View Books
3. Search Book
4. Remove Book
5. Exit
Enter your choice: 3
Enter book name to search:Python
Book not found.
===== Library Management System =====
1. Add Book
2. View Books
3. Search Book
4. Remove Book
5. Exit
Enter your choice: 4
Enter book name to remove:Programming
Book not found
===== Library Management System =====
1. Add Book
2. View Books
3. Search Book
4. Remove Book
5. Exit
Enter your choice:
Launchpad
Ln 49, Col 9 Spaces: 4 UTF-8 CRLF Python Sign In Python 3.14.3 Go Live
24°C
Mostly cloudy
Search
ENG
IN
01:55
09-08-2026